# RayCastED: A RayCast End-to-end Detection Model for Lightweight and Boundary-Aware Nuclei Detection

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# Clinical Background: TILs Assessment

- **Tumor-infiltrating lymphocytes (TILs)** are established prognostic biomarkers in multiple cancers
- International TILs Working Group guidelines recommend **area-based scoring**: percentage of stromal area occupied by lymphocytes
- **Critical requirement**: accurate boundary delineation, not just centroid detection

**The Challenge**

Manual assessment is:

- Time-consuming
- Subject to inter-observer variability
- Impractical for whole-slide images

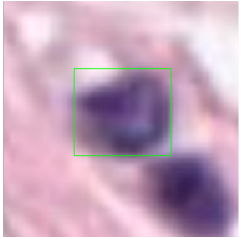
# The Clumped Cell Problem

- Densely clustered nuclei merge in segmentation masks
- FCN-based models produce contiguous predictions
- **Result**: Systematic undercounting
- Existing solutions require costly post-processing
- NMS suppresses overlapping predictions (risky in clumped regions)

**Key Issues**

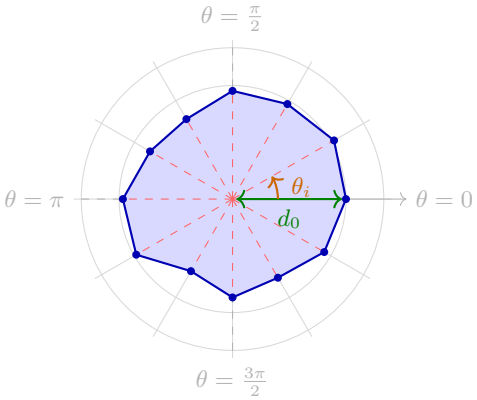
1. Instance separation overhead
2. NMS-based suppression errors
3. Deployment inaccessibility (transformers)

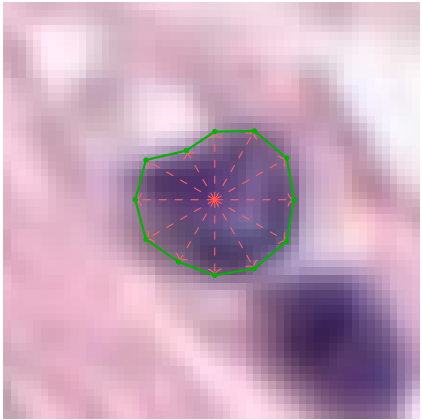
# The Accuracy-Deployment Trade-off



| **Approach** | **Pros** | **Cons** |
|---|---|---|
| Instance Segmentation (U-Net, Mask R-CNN) | Accurate boundaries | Computationally expensive (unsuitable for edge) |
| Transformer-based (CellViT, DINO) | Strong performance<br>Good separation | Quadratic complexity (memory/compute intensive) |
| Object Detection (YOLO, Faster R-CNN) | Fast, lightweight<br>Edge-deployable | **No boundary geometry** (insufficient for TILs) |

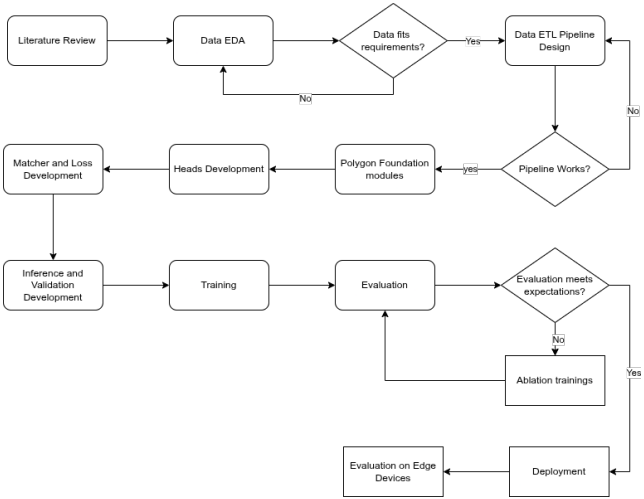
# Star-Convex Polygons for Cell Representation



- **Left**: Polar coordinate representation with centroid $\mathbf{c}$ and radial distances $d_i$
- **Right**: Applied to real cell with 12 rays at fixed angles $\theta_i$
- Preserves boundary geometry while maintaining per-cell separation

# Research Workflow



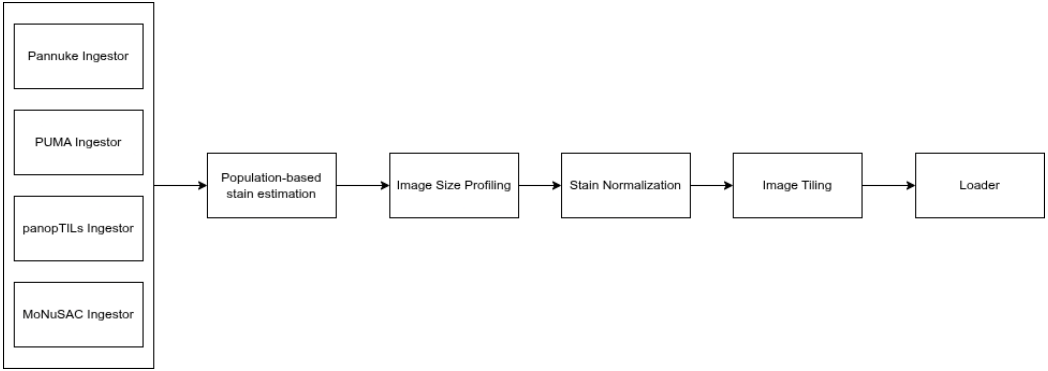
# Datasets

| Dataset | Purpose | Tissue | Format |
|---|---|---|---|
| PanNuke | Primary benchmarking | Multi-organ | Pickled bytes |
| PUMA | TILs detection | Melanoma | TIFF, GeoJSON |
| panopTILs | TILs detection | Breast | PNG |
| MoNuSAC | TILs detection | Multi-organ | Pickled bytes |

Table: Summary of datasets used in this study.

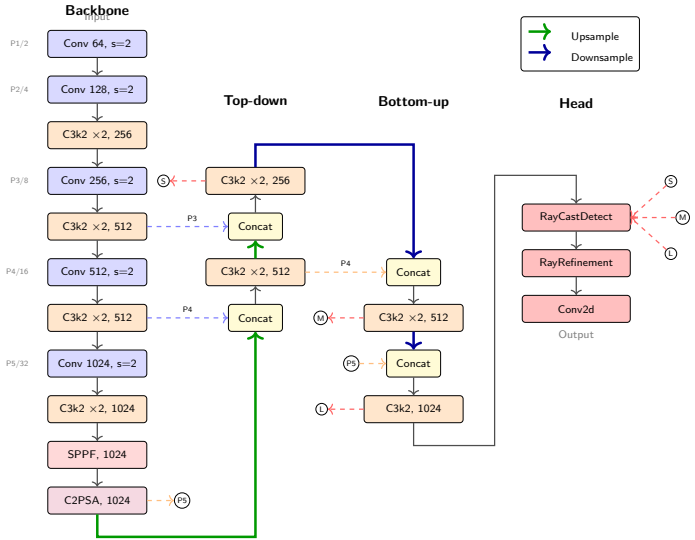
- H&E stained histopathology images
- $\sim$0.25 MPP resolution
- Multiple tissue types for generalization

# Data Preprocessing Pipeline




- Multi-format dataset handling (TIFF, PNG, pickled bytes)
- Patch extraction and augmentation
- Star-convex annotation generation

# RayCastED Architecture



# Key Innovations

## 1. RayCast Representation

- Centroid $(x, y)$ + 12 radial distances
- Fixed-angle rays $(0, 30, ..., 330)$
- End-to-end differentiable

## 2. End-to-End Assignment

- Learned bipartite matching
- Eliminates NMS dependency
- Prevents undercounting in clumps

## 3. Lightweight FCN Backbone

Replaces transformer with YOLO-based FCN for edge deployment (TensorRT/ONNX Runtime)

# Evaluation Against LSP-DETR

| Metric | LSP-DETR | RayCastED | Delta |
|---|---|---|---|
| AJI | **0.825** | 0.6385 | -0.1865 |
| FI | **0.677** | 0.5581 | -0.1189 |
| mAP@0.5 | **0.691** | 0.4275 | -0.2635 |
| Precision | **0.862** | 0.6848 | -0.1772 |
| Recall | **0.791** | 0.5980 | -0.1930 |
| PQ | **0.657** | 0.5433 | -0.1137 |
| SQ | **0.811** | 0.7460 | -0.0650 |
| DQ | **0.810** | 0.6817 | -0.1283 |
| Params | 45.0M | **11.92M** | **-73.5%** |
| GFLOPs | - | 4.77 | - |
| Inference | - | **4.17ms** | - |

Table: Performance comparison with LSP-DETR baseline.

RayCastED achieves **73.5% fewer parameters** while maintaining competitive segmentation quality, enabling edge deployment.